

Cigarette Smoke Toxins Deposited on Surfaces: Implications for Human Health

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Cigarette smoking remains a significant health threat for smokers and nonsmokers alike. Secondhand smoke (SHS) is intrinsically more toxic than directly inhaled smoke. Recently, a new threat has been discovered – Thirdhand smoke (THS) – the accumulation of SHS on surfaces that ages with time, becoming progressively more toxic. THS is a potential health threat to children, spouses of smokers and workers in environments where smoking is or has been allowed. The goal of this study is to investigate the effects of THS on liver, lung, skin healing, and behavior, using an animal model exposed to THS under conditions that mimic exposure of humans. THS-exposed mice show alterations in multiple organ systems and excrete levels of NNAL (a tobacco-specific carcinogen biomarker) similar to those found in children exposed to SHS (and consequently to THS). In liver, THS leads to increased lipid levels and non-alcoholic fatty liver disease, a precursor to cirrhosis and cancer and a potential contributor to cardiovascular disease. In lung, THS stimulates excess collagen production and high levels of inflammatory cytokines, suggesting propensity for fibrosis with implications for inflammation-induced diseases such as chronic obstructive pulmonary disease and asthma. In wounded skin, healing in THS-exposed mice has many characteristics of the poor healing of surgical incisions observed in human smokers. Lastly, behavioral tests show that THS-exposed mice become hyperactive. The latter data, combined with emerging associated behavioral problems in children exposed to SHS/THS, suggest that, with prolonged exposure, they may be at significant risk for developing more severe neurological disorders. These results provide a basis for studies on the toxic effects of THS in humans and inform potential regulatory policies to prevent involuntary exposure to THS.